

Iterative Progressive Retrieval Attention: Bounded Associative Closure over Latent Gist Memory

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Abstract

One-shot semantic retrieval finds memory nearest to the current query. It can miss a second piece of evidence that is weakly related to the query but strongly related to the first piece. We study iterative Progressive Retrieval Attention (PRA), which treats cached routing gists as an implicit graph: the query retrieves a bounded frontier, retrieved gists query the same layer-local index, and only the final closure is mapped to full native key/value (K/V) state. The transformer is not rerun between retrieval rounds.

A parameter-free implementation enforces depth, branch, beam, and unique-parent budgets and emits a retrieval graph before K/V materialization. Controlled two- and three-hop graphs validate chunk closure exactly, but inherited natural-data results are mixed. We therefore test two specific failure modes without retraining. First, the learned router is asymmetric: $z_q = W_q h_q$ and $z_m = W_m h_m$. Replacing the incorrect hop-2 frontier $W_m h_A$ by $W_q h_A$ recovers 0.075 QASPER any-evidence recall relative to historical closure at 10%, yet remains 0.013 below one-shot; HotpotQA chain completion is unchanged. Second, we encode 256-token parents once, derive eight 32-token local gists from each contextual state, traverse local nodes, and deduplicate parents before materializing native K/V. This local method solves every designed bridge control, where parent closure always fails. Finally, semantic narrowing followed by tokenwise layer-27 pre-RoPE query-key (QK) matching solves all 320 native-bridge controls. On natural data at a matched 20% parent budget, however, the primary native-QK condition reaches HotpotQA chain completion 0.125 versus 0.125 local semantic and 0.138 one-shot; QASPER reaches 0.900 versus 0.925 and 0.938. A Top-4 control reaches 0.175 on HotpotQA but falls to 0.838 on QASPER and does not dominate across budgets. Native propagation is therefore expressively sufficient on designed bridges but not a dependable pretrained associative geometry on these natural samples. One-shot remains the default, and any learned associative extension now has a clearer justification and baseline. An exact-oracle audit finds that iterative native routing loses 11.7–15.6% of QASPER runs where one-shot already touched oracle evidence. An oracle-protected-root counterfactual recovers 0.044–0.068 recall at the same budget, whereas held-out score normalization does not repair HotpotQA. Monotonic root preservation, then adaptive transition width, are therefore tested as a final frozen-policy gate. On the held-out split, monotonic preservation exactly reproduces one-shot quality. An exploratory native-rank policy raises 20% Hotpot oracle recall from 0.205 to 0.233 but leaves chain completion unchanged at 0.143 and regresses at larger budgets; QASPER is preserved at 20%. The first Hotpot evidence group is absent from root Top- B in 52.9% of exploratory runs, compared with 11.1% where it is present but not locked. We therefore retain the full contextual question once and expose overlapping query facets to one shared Top- B competition. Validation selects 4-token windows at stride 2. At 20%, they raise held-out Hotpot entry presence from 0.343 to 0.457, MRR from 0.503 to 0.626, and oracle recall from 0.205 to 0.295, while QASPER entry presence remains 0.967. Real pre-RoPE head splitting does not improve this result. Reconnecting the faceted root to the frozen monotonic loop raises Hotpot chain completion from 0.143 to 0.171, but no higher than

the faceted root alone. The supported extension is thus opt-in multi-span entry search, not a new default or a claim that propagation is solved.

1 Introduction

Suppose a question is directly related to memory chunk A . Chunk A , in turn, identifies an entity or relation needed to find chunk B . A one-shot router computes similarities from the question to all chunks. It can retrieve A while missing B because the question does not yet contain the intermediate relation. Increasing top- k may find B , but it spends the same extra budget on unrelated chunks. The problem is not transformer reasoning after evidence arrives. It is evidence discovery before full memory becomes active.

PRA already separates compact semantic routing state from full layer-native K/V payloads [8, 7, 9]. This separation suggests a narrow extension. After retrieving A , use A 's compact gist as another query. Continue for a bounded number of rounds, preserve the discovery graph, and materialize exact K/V only once. The resulting path $q \rightarrow A \rightarrow B$ is explicit. It does not require another language-model pass, an intermediate text answer, or serialization through an external tool API.

This paper asks whether pretrained hidden-state gists already possess enough transitive semantic geometry for that extension to help. The question has a strict baseline: iterative closure must beat one-shot Top- B at the same final unique-chunk budget B . Otherwise, retrieval depth adds cost without adding a useful primitive.

We keep the pretrained backbone and Paper 2 routing/transport stack fixed and replace one-shot Top- B discovery with parameter-free iterative closure. Projection correction, local-gist propagation, and native-QK propagation likewise alter discovery state only; they do not train the backbone or a memory-consumption adapter.

The contributions are eightfold. First, we formulate iterative PRA as query-conditioned bounded associative closure over an implicit latent graph. Second, we implement Level-1 gist-to-gist and Level-2 aggregate-frontier traversal with deterministic budgets, cycle handling, root anchoring, transparent path scores, and a versioned graph artifact. Third, we evaluate five semantic-router seeds on held-out HotpotQA and QASPER features, together with exact two- and three-hop synthetic controls. Fourth, we identify and correct an asymmetric projection-interface error without retraining. Fifth, we separate contextual encoding, local propagation, and parent K/V materialization scales, showing exact local-bridge recovery but no dependable natural-data chain gain. Sixth, we test semantically narrowed, tokenwise pre-RoPE native QK propagation under the same final parent budget, separate semantic and native search costs, and diagnose exact-oracle convergence, root displacement, edge rank, and score calibration before choosing the next gate. Seventh, we implement oracle-free monotonic root locks and bounded adaptive transition competition, select their thresholds on an example-disjoint validation split, and show on held-out data that persistence repairs direct-retrieval displacement but does not create a Hotpot chain gain when the entry evidence never enters root Top- B . Eighth, we isolate query-span and native head dilution, show that contextual query facets materially improve held-out Hotpot entry discovery without reducing QASPER recall, and reconnect the winning root to the frozen bounded propagation loop under an unchanged final payload budget.

2 From Nearest Neighbors to Associative Closure

2.1 Why one-shot retrieval can stop too early

Let $G = \{g_1, \dots, g_N\}$ be compact routing gists at one transformer layer and let q be the current routing query. One-shot PRA returns

$$N_B(q) = \text{TopB}_{g \in G} s(q, g), \quad (1)$$

where s is cosine similarity in the learned semantic space. This operation is appropriate when every useful chunk is independently similar to q . It is incomplete when

$$s(q, g_A) \gg 0, \quad s(q, g_B) \approx 0, \quad s(g_A, g_B) \gg 0. \quad (2)$$

Here A is an entry point and B is relationally accessible only after A is known.

We interpret each chunk as a node in an implicit graph. High gist similarity supplies candidate directed edges, generated lazily when a node becomes part of the frontier. No $N \times N$ adjacency matrix is built. Iterative PRA therefore computes a bounded semantic neighborhood, not a proof and not complete reasoning.

2.2 The complete data path

Figure 1 separates traversal from payload activation. This boundary is the main systems invariant: compact gists may participate in several search rounds, while full K/V cannot enter attention until the closure stops.

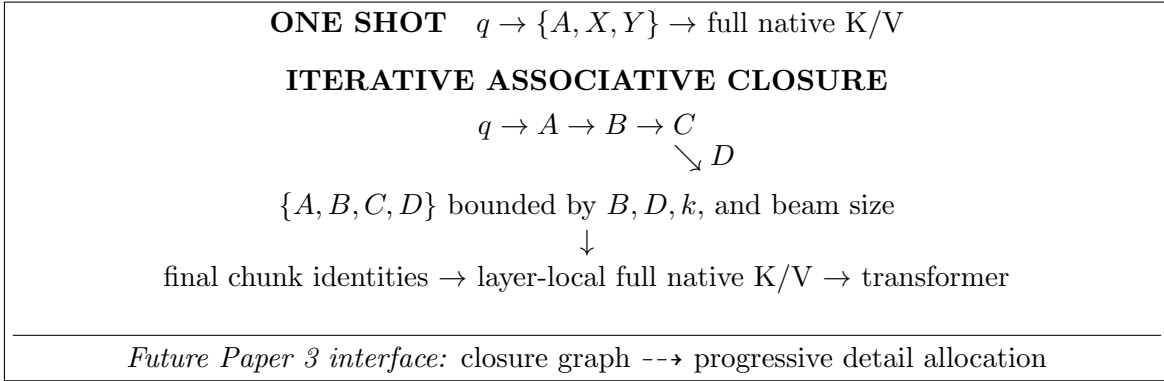


Figure 1: Iterative PRA increases search depth without transformer reruns. Gists are routing state; native K/V is the payload. The dashed handoff preserves graph signals for later work on partial materialization.

3 Method

3.1 Bounded frontier expansion

The root query initializes frontier $F_0 = \{q\}$ and the visited set is empty. At round t , each frontier vector proposes its k highest-scoring unseen chunks. Duplicate proposals are merged, the best path to each candidate is retained, and at most the remaining beam and unique-node budgets are accepted. Accepted gists form F_t . Traversal stops at fixed depth D , final budget B , no-new-node convergence, or an optional confidence threshold.

To limit semantic drift, candidate c is scored against both the root and its frontier parent:

$$e_t(c) = \alpha s(q, g_c) + (1 - \alpha)s(f_{t-1}, g_c), \quad 0 \leq \alpha \leq 1. \quad (3)$$

At $\alpha = 0$, traversal follows local gist edges. At $\alpha = 1$, every round reproduces root ranking after visited-node removal. Intermediate values preserve the original information need while allowing a retrieved node to redirect search.

The primary path score multiplies affinities $a_i = (e_i + 1)/2$ along a path:

$$P(c_t) = \prod_{i=1}^t a_i. \quad (4)$$

We also evaluate log-sum, final-edge, minimum-edge, mean-edge, and direct-root scores. These choices are deliberately transparent. A learned controller would confound the first question: whether useful graph structure is already present.

Listing 1: Level-1 bounded associative closure

```
frontier = {root}; selected = {}; graph = empty
for hop in 1..D:
    proposals = union topk_unseen(parent, G, k) for parent in frontier
    merge duplicate targets; retain parent edges and best path
    accepted = top(proposals, min(beam, B - len(selected)))
    add accepted nodes and edges to graph
    frontier = winning_gists(accepted)
    stop if frontier is empty or len(selected) == B
materialize_full_native_kv(selected) # only after closure
```

3.2 Level-2 frontier updates

Direct Level-1 traversal uses each winning gist as the next query. We compare three parameter-free updates. A residual update keeps parent state,

$$r_{t+1} = \text{normalize}(r_t + \beta g_t), \quad (5)$$

while mean and path-weighted mean collapse accepted branches to one aggregate vector and add the root. These updates test whether mild state accumulation reduces drift. They do not rerun a transformer and introduce no learned parameters.

3.3 Asymmetric projection contract

Paper 2’s frozen router does not embed questions and memories with one shared map. It computes

$$z_q = W_q h_q, \quad z_m = W_m h_m, \quad (6)$$

and scores $s(q, m) = \cos(z_q, z_m)$. The original closure stored z_m and reused it as the next frontier. Its second hop was $\cos(W_m h_A, W_m h_B)$, although the learned interface expects

$$s(A, B) = \cos(W_q h_A, W_m h_B). \quad (7)$$

Equal tensor width does not make the two projections semantically interchangeable. The corrected cache derives and stores an aligned query-projected companion gist when it builds the memory index. No parameter changes and no retraining are involved. Root ranking remains $W_q h_q \rightarrow W_m h_B$; only hop 2 and later use $W_q h_A$.

3.4 Contextual parents and local propagation

Chunk means can hide a narrow relational bridge. We therefore distinguish three scales:

$$\boxed{\text{encoding granularity} \neq \text{propagation granularity} \neq \text{materialization granularity.}} \quad (8)$$

The fixed baseline encodes a 256-token parent once through Qwen. It divides that already contextualized hidden-state tensor into eight contiguous 32-token windows and mean-pools one local gist per window. Local windows are never independently re-encoded. A node has identity (parent_id, local_id) and carries its source span and both W_q and W_m projections. Traversal is

$$q \rightarrow g_{A,i} \rightarrow g_{B,j}, \quad s(A_i, B_j) = \cos(W_q h_{A,i}, W_m h_{B,j}). \quad (9)$$

Candidate local nodes are aggregated by parent, previously selected parents are removed, and the best local activation proposes each new parent at most once. Final selection materializes whole 256-token parent K/V under the same parent-count budget as one-shot and parent closure. The experiment records local nodes explored, cross-parent transitions, local and parent entropy, and

$$\Delta_{bridge} = s(g_{A,i}, g_{B,j}) - s(g_A, g_B). \quad (10)$$

3.5 Budgets, cycles, and layer locality

Four controls bound expansion: depth D , neighbors per frontier node k , per-round beam size, and maximum unique chunks B . A visited set prevents $A \rightarrow B \rightarrow A$ from consuming budget twice. Alternate parents remain graph edges even when one path wins pruning. Every index is layer-local. A query, gist, and native payload from different layers are never treated as interchangeable merely because their dimensions match.

If C is the number of chunks and $|F_t|$ the frontier size, Level-1 scoring performs $C \sum_t |F_t|$ gist comparisons. Full K/V transfer remains proportional to the final selected payload, exactly as in matched one-shot PRA. The method trades more compact-index work for a chance to find a better closure without increasing active K/V.

4 Experimental Design

4.1 Inherited Paper 2 configuration

Table 1 makes the inherited stack explicit. Layer numbers are zero based. The retrieval studies operate on frozen feature artifacts and do not run the transformer or activate K/V; the separate downstream smoke is the only Paper 2.5 experiment that exercises the complete native-K/V consumer path.

Table 1: Verified inherited configuration and Paper 2.5 experimental scope.

Component	Paper 2.5 setting
Backbone	Qwen/Qwen3-0.6B, frozen; 596,049,920 parameters and no trainable backbone parameter during feature capture.
Backbone revision	Hugging Face commit <code>c1899de289a04d12100db370d81485cdf75e47ca</code> . The inherited feature manifest and downstream smoke pin this hash. The Gate-2 manifest recorded the alias <code>main</code> ; the cached ref used for that run resolves to the same hash.
Backbone topology	28 decoder layers (zero-based 0–27), hidden width 1,024, 16 query heads, 8 K/V heads, and native head width 128.
Routing source	Attention-input hidden states from decoder layer 27; the original root uses the final prompt-token hidden state. The entry-facet gate retains that exact row and adds means over overlapping question-token states from the same full prompt pass.
Contextual encoding and routing unit	The inherited feature artifact encodes contiguous 128-token blocks and exposes 32-token routing views. These views are derived from contextual hidden states, not independently encoded fragments. Gate 2 separately encodes 256-token parents once and derives eight 32-token local means from each parent state.
Raw routing gist	One mean hidden-state gist per inherited 32-token routing chunk. Gate 2 retains one local mean per 32-token subwindow and an occupancy-weighted 256-token parent mean.
Native closure representation	Gate 3 uses tokenwise layer-27 pre-RoPE Q/K inside contextual 32-token regions, with Qwen-compatible 16-query-head to 8-K/V-head pairing. These tensors are routing only; the materialized payload remains post-RoPE native K plus native V.
Learned discovery component	The only inherited learned discovery component is Paper 2’s asymmetric linear semantic router. It maps a 1,024-D hidden state to a 128-D routing space with 262,144 parameters.
Router projections	$z_q = W_q h_q$ and $z_m = W_m h_m$; one-shot scoring is $\cos(z_q, z_m)$.
Router checkpoints	Five independently trained exhaustive-negative, margin-objective checkpoints with seeds 11, 23, 37, 53, and 71. No checkpoint is trained or selected on Paper 2.5 iterative results.

Table 1 (continued).

Component	Paper 2.5 setting
Iterative closure	Parameter-free. The primary chunk study uses direct frontiers, multiplicative path score, root anchor $\alpha = 0.25$, and depths 2, 3, and 4; one-shot is the matched depth-1 Top- B reference. Gate 3 also remains parameter-free.
Memory-consumption adaptation	None. Paper 2.5 does not load a memory-use adapter. Residual-16, rank-32 LoRA, residual-plus-LoRA, and global LM-head adaptation are all absent.
PRA payload	Full layer-native post-RoPE K and native V, preserving Qwen’s 8 K/V heads without expanding them to 16 query heads.
PRA consumer layers	None in the frozen-feature retrieval experiments. The downstream integration smoke routes at layer 27 and materializes/consumes native K/V on layer 27 only; it does not use a multi-layer late band.
Routing versus materialization	Intermediate closure nodes remain compact routing state and never enter attention. Only final selected identities are handed to the existing layer mapper before native K/V becomes active.
Native-operation bounds	The configured hard bound is 256 tokens. Feature encoding uses 128-token blocks in the inherited study and 256-token parents in Gate 2. The downstream smoke uses 64-token encoding blocks, observes a maximum native operation of 64 tokens, and records zero violations.
Final native payload budget	The downstream smoke caps materialized memory at 96 tokens and selects three 32-token chunks in both one-shot and iterative conditions. Retrieval-only studies apply matched logical chunk/parent budgets but do not transfer K/V.
K/V residency in smoke	Reference K/V is CPU-resident, pinned when CUDA is available, and transferred non-blockingly for final selected identities.
Datasets	A held-out split of 16 HotpotQA and 16 QASPER examples, plus controlled synthetic two-hop, three-hop, and local-bridge tasks.
New Paper 2.5 variables	Discovery procedure only: one-shot versus bounded iterative closure, followed by projection-correct frontier, parent-versus-local propagation, semantically narrowed native-QK propagation, monotonic competition, and contextual query-facet gates. Backbone weights, router checkpoints, payload representation, layer mapper, and host attention semantics remain fixed within each matched comparison.

The resulting path is

frozen Qwen hidden state $\rightarrow$ learned 128-D asymmetric router $\rightarrow$ one-shot or iterative closure $\rightarrow$ final identities $\rightarrow$ PRA layer mapper $\rightarrow$ native K/V on layer 27.

Iterative closure operates only over compact routing state. Intermediate graph nodes do not inject K/V into the transformer. After closure terminates, the final selected identities are handed to the existing PRA layer mapper, which materializes the corresponding full native K/V payload on the configured consumer layer in the downstream smoke.

Paper 2.5 changes the discovery procedure only. The pretrained backbone, routing checkpoints, native-K/V payload representation, layer mapper, and host-model attention semantics remain fixed within each comparison. Therefore differences between one-shot and iterative conditions are routing/closure effects, not effects of additional model training or a different memory-consumption adapter.

Table 2: Isolation of inherited components from Paper 2.5 contributions.

Inherited from Paper 2	New in Paper 2.5
Frozen pinned Qwen3-0.6B	Bounded iterative closure and retrieval graph
128-D asymmetric semantic router	Depth, path, root-anchor, and stop logic
Attention-input hidden-state gists	Projection-correct frontier audit
Native post-RoPE K/V transport and mapper	Hierarchical local propagation with parent deduplication
Bounded native execution	Matched-budget chain metrics and transitive controls
Fixed post-RoPE native K/V payload	Pre-RoPE native-QK propagation audit
Final-token global query	Contextual multi-span query entry audit

4.2 Frozen semantic geometry

We reuse Paper 2’s frozen Qwen3-0.6B feature artifact and its attention-input hidden-state representation [5, 9]. The held-out split contains 16 HotpotQA and 16 QASPER examples, 32-token chunks, exact evidence-overlap masks, and 1,024-dimensional root and memory features [11, 2]. Five asymmetric linear projections, each with 262,144 parameters and seeds 11, 23, 37, 53, and 71, map query and memory features to 128 dimensions. They were trained on the disjoint Paper 2 train split with exhaustive negatives. No router is trained or selected on iterative results.

The primary setting uses direct frontier gists, multiplicative path score, $\alpha = 0.25$, and $D \in \{2, 3, 4\}$. Final budgets are 5, 10, 20, and 30% of each example’s candidate chunks. One-shot always receives the identical final B . We separately sweep $\alpha \in \{0, .25, .5, .75, 1\}$, all Level-2 updates, and all six path scores at $D = 2$, 10%.

4.3 What the evidence metrics mean

Any-evidence recall asks whether one annotated chunk is selected. Exact all-evidence recall asks whether every evidence-overlapping 32-token chunk is selected. Because one supporting sentence can cross a chunk boundary, we additionally group contiguous positive chunks and define chain completion as selecting at least one chunk from every group. The artifact reports budget feasibility for both definitions. Hotpot groups are ordered by source position; they are not claimed to encode causal hop order. QASPER usually has one contiguous evidence group and is therefore a direct-retrieval control.

The synthetic controls provide exact causal chains. Their root is similar to node A ; each next node is similar to its predecessor but is outranked from the root by distractors. At matched $B = 2$ or 3, one-shot can afford the entry node but not the full chain. We run 64 examples per seed for each chain length, yielding 320 paired trials per task.

4.4 Downstream execution validation

A separate two-example CUDA smoke loads the pinned Qwen model, indexes one HotpotQA and one QASPER source, and compares no memory, one-shot, and $D = 2$ closure. One-shot and closure both request three chunks and materialize 96 native K/V tokens. This run validates execution and budget invariants; two examples cannot estimate answer quality.

4.5 Gate-3 native-QK protocol

Gate 3 asks whether initial relevance and associative propagation require different pretrained geometries. The learned semantic router still scores the root and narrows each activated local frontier to either 10% or 20% of candidate parents. Only local regions inside those parents enter native refinement. From the same layer-27 contextual states used by Gate 2, we retain Qwen’s normalized pre-RoPE projections

$$Q^{pre} = W_Q h, \quad K^{pre} = W_K h. \quad (11)$$

Qwen has 16 query heads and 8 K/V heads, so query heads $2j$ and $2j + 1$ compare only with K/V head j . The primary local-region transition score is

$$s_{QK}(A_i, B_j) = \max_{a \in A_i, b \in B_j, h} \frac{q_{a,h}^T k_{b, \lfloor h/2 \rfloor}}{\sqrt{128}}. \quad (12)$$

The sole smooth control averages the four strongest valid token pairs per head and then the four strongest query heads. We compare fixed Top- k transitions with one threshold, $\theta = \mu + \sigma$, always under the same hard parent cap. If thresholding accepts too few parents, root-semantic ranking fills the remainder so final materialization stays matched.

Each 256-token parent is encoded once. Its token states are sliced into contextual 32-token routing regions; no region is independently re-encoded. Native QK changes discovery identities only. Intermediate states contain no payload K/V, and final identities would still map through the existing post-RoPE native K plus native V path. The five router seeds, 10/20/30% final parent budgets, and all one-shot, parent-closure, and local-semantic baselines remain fixed. These Gate-3 baselines are recaptured on the same strict 32-token grid as native QK. They are internally matched but slightly differ from Gate 2’s historical eight-equal-segment artifact when a final 256-token parent is short; Tables 6 and 7 therefore answer their respective within-gate comparisons rather than being pooled.

5 Results

5.1 Exact latent chains validate the mechanism

Table 3 gives the cleanest mechanism test. Both methods find some evidence, but one-shot retrieves only the entry node. Iterative traversal follows the designed edges and completes every chain. The gain costs one complete index scan per hop.

Table 3: Matched-budget synthetic results over five seeds and 64 examples per seed.

Task	Method	Any	Exact chain	Coverage	Comparisons
2-hop	One-shot Top-2	1.000	0.000	0.500	10
2-hop	Iterative $D = 2$	1.000	1.000	1.000	20
3-hop	One-shot Top-3	1.000	0.000	0.333	11
3-hop	Iterative $D = 3$	1.000	1.000	1.000	33

This result establishes that the implementation can recover indirect nodes without transformer updates. It does not show that a pretrained semantic router supplies the required edges.

5.2 HotpotQA shows a small chain signal and a larger drift cost

Figure 2 and Table 4 compare natural-data routing. At 10%, one-shot any-evidence recall is 0.700 ± 0.065 (five-seed 95% interval), while $D = 2$ obtains 0.663 ± 0.118 . Yet group-level chain completion rises from 0.138 to 0.150, and reaches 0.175 at $D = 4$. The two metrics describe different behavior: closure sometimes reaches another evidence group, but also leaves an easily retrieved entry neighborhood.

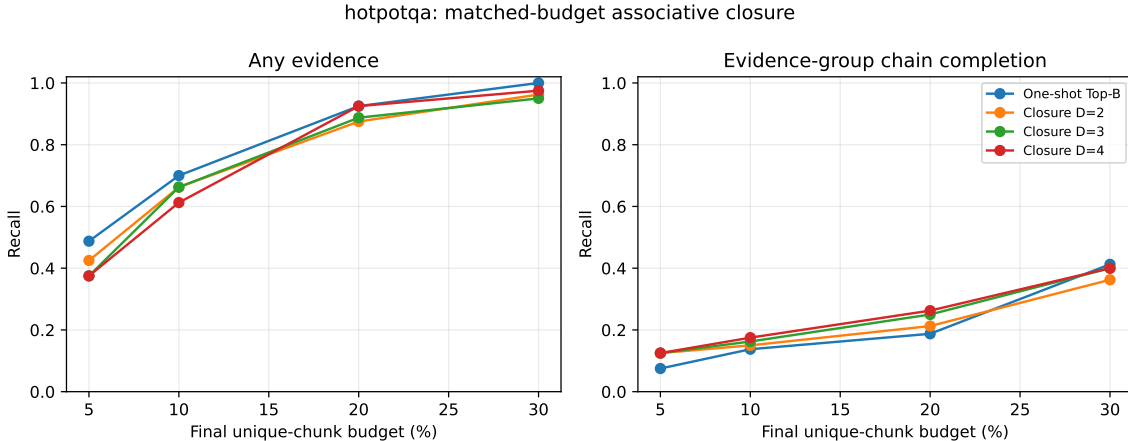


Figure 2: Five-seed HotpotQA matched-budget curves. Deeper closure modestly improves chain completion at some sparse budgets but does not improve the broader any-evidence curve.

Table 4: Primary 10% matched-budget results. Means combine 16 examples and five independently trained routers. “All” requires every positive chunk; “Chain” requires every evidence group.

Dataset	Method	Any	Chain	All	Coverage	Comparisons
HotpotQA	One-shot	.700	.138	.013	.207	50.1
HotpotQA	Closure $D = 2$	.663	.150	.000	.202	219.6
HotpotQA	Closure $D = 3$	.663	.163	.000	.205	273.0
HotpotQA	Closure $D = 4$	.613	.175	.000	.193	303.2
QASPER	One-shot	.963	.963	.213	.477	142.6
QASPER	Closure $D = 2$	.875	.875	.200	.466	1395.9
QASPER	Closure $D = 3$	.788	.788	.175	.425	1822.1
QASPER	Closure $D = 4$	.825	.825	.150	.426	2144.2

At 20%, Hotpot chain completion rises from 0.188 one-shot to 0.212, 0.250, and 0.263 at depths 2–4. At 30%, one-shot returns to the lead (0.412 versus 0.362–0.400). The effect is therefore small, budget dependent, and not accompanied by improved exact all-chunk recall. Five seeds are insufficient to claim a reliable improvement; the direction also changes with budget.

5.3 QASPER behaves as a direct-retrieval control

QASPER’s evidence commonly forms one contiguous group, so chain completion equals any-evidence recall in this protocol. Figure 3 shows that one-shot dominates throughout the sparse

region. At 10%, its any/chain recall is 0.963 ± 0.069 versus 0.875 ± 0.095 for $D = 2$. Closure adds no useful intermediate relation when the root already addresses the target.

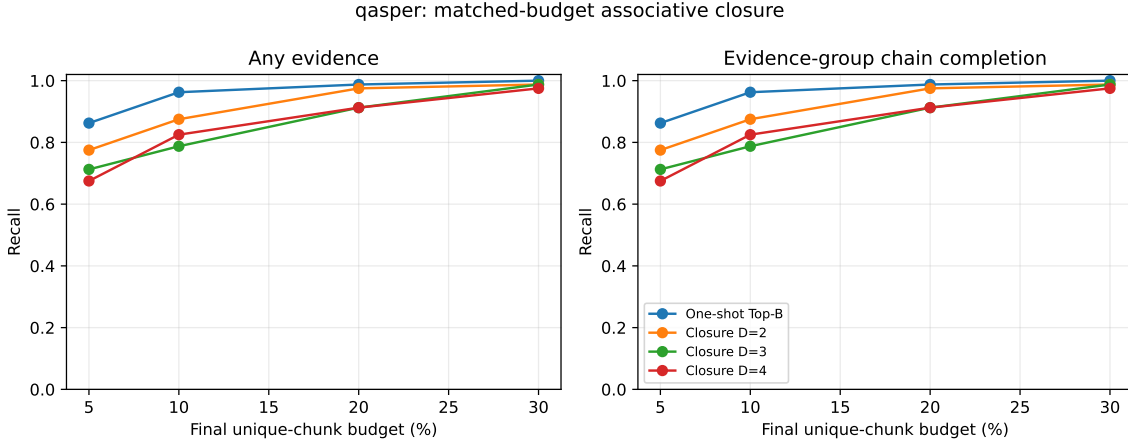


Figure 3: QASPER matched-budget curves. The mostly single-group evidence protocol favors direct one-shot ranking, as expected if recursion is unnecessary.

5.4 Root anchoring exposes semantic drift

At 10% Hotpot, pure local traversal ($\alpha = 0$) yields any recall 0.600. Increasing root weight to 0.50, 0.75, and 1.0 gives 0.700, 0.738, and 0.700. QASPER improves monotonically toward the root-only condition: 0.863, 0.875, 0.900, 0.913, and 0.963 from $\alpha = 0$ to 1. The best QASPER “iterative” setting is therefore the one that does not use frontier semantics.

Residual update is the only Level-2 variant to preserve exact all-chunk recall on Hotpot (0.013), but its any recall 0.713 does not consistently beat one-shot. Mean and weighted mean reach 0.738 any recall while exact all remains zero. Path-score variants range from 0.650 to 0.738 Hotpot any recall and do not improve chain completion consistently. No parameter-free update dominates.

5.5 Relational gap does not identify real gains

The motivating diagnostic is

$$\Delta_{rel} = \max_{A \in E \setminus \{B\}} s(g_A, g_B) - s(q, g_B). \quad (13)$$

Large Δ_{rel} should favor closure if evidence-to-evidence similarity is a useful edge. Figure 4 does not show that pattern. In Hotpot’s highest gap quartile, the paired $D = 2$ any-recall effect is -0.100 and coverage effect is $+0.017$. In QASPER they are -0.176 and -0.019 . A high pairwise cosine score is not enough to identify the correct relational transition.

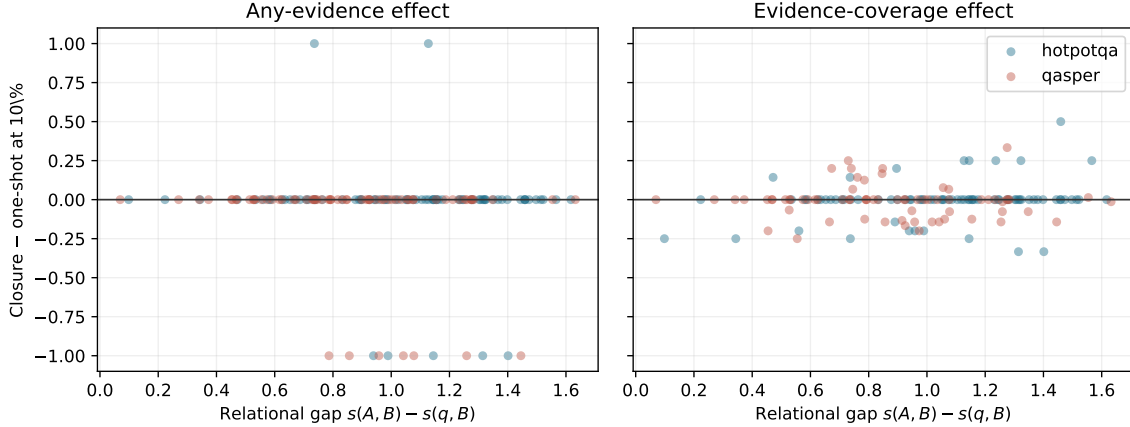


Figure 4: Paired $D = 2$ minus one-shot effects at 10%. Relational gap does not predict a stable closure advantage in the inherited semantic space.

5.6 Search cost and full-K/V execution

Figure 5 shows the compact-index cost. At 10%, $D = 2$ uses 219.6 comparisons per Hotpot example versus 50.1 one-shot ($4.4\times$), and 1395.9 versus 142.6 on QASPER ($9.8\times$). The higher QASPER multiplier follows from more candidates and larger frontiers. These are exact comparison counts; wall time depends strongly on batching and hardware.

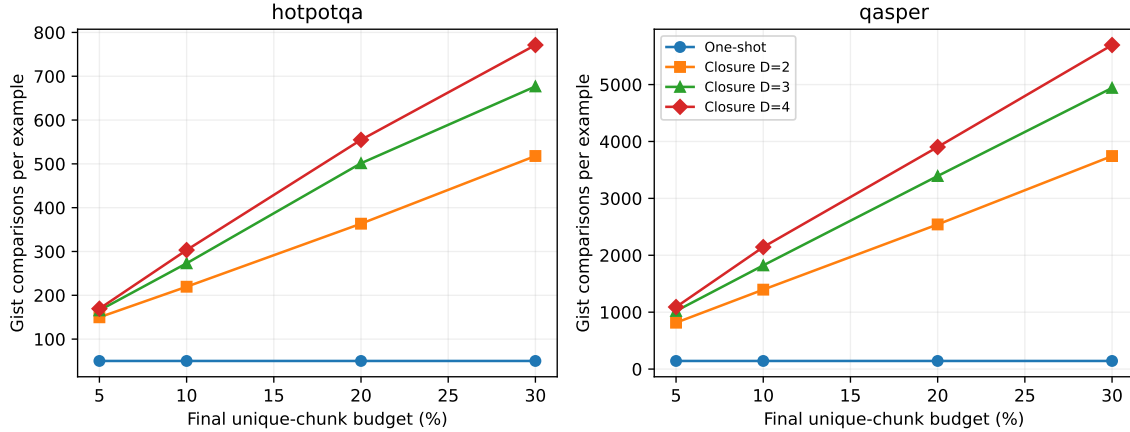


Figure 5: Compact gist comparisons. Iteration leaves final K/V budgets fixed but increases index work with frontier size and depth.

The CUDA integration smoke materializes exactly three chunks and 96 K/V tokens for both methods. Iterative routing takes 0.103/0.077 seconds on the Hotpot/QASPER examples, versus 0.107/0.073 for one-shot in this trace-oriented implementation. End-to-end times are 1.291/1.291 seconds for closure and 1.304/1.347 seconds for one-shot. The model records zero native-limit violations and a maximum 64-token native operation. These two cases validate the execution contract only.

6 Projection and Locality Gates

The preceding results freeze the original chunk-level closure as Level 1. We next change one assumption at a time. All comparisons reuse the same five checkpoints, held-out examples, depth, root anchor, path score, and final unique-parent/K/V budget. Neither gate trains a new parameter.

6.1 Gate 1: projection-correct frontiers

Table 5 compares the historical $W_m h_A$ frontier with the contract-correct $W_q h_A$ frontier at the primary 10% chunk budget. Correction materially reduces the QASPER failure: any/chain recall rises by 0.075 and coverage by 0.013 relative to historical closure. It nevertheless remains 0.013 below one-shot any/chain recall. On HotpotQA, correction raises any recall by 0.025 and restores exact identity by 0.013 relative to historical closure, but chain completion falls by 0.013 and exactly matches one-shot. Comparison counts are unchanged because this gate changes only the projection used by each activated frontier.

Table 5: Gate 1 at a matched 10% final chunk budget. Q-frontier is the corrected $W_q h_A \rightarrow W_m h_B$ traversal; M-frontier is the historical direct reuse of $W_m h_A$.

Dataset	Method	Any	Chain	Exact	Coverage	Comparisons
HotpotQA	One-shot	.700	.138	.013	.207	50.1
HotpotQA	M-frontier	.663	.150	.000	.202	219.6
HotpotQA	Q-frontier	.688	.138	.013	.201	219.6
QASPER	One-shot	.963	.963	.213	.477	142.6
QASPER	M-frontier	.875	.875	.200	.466	1395.9
QASPER	Q-frontier	.950	.950	.225	.479	1395.9

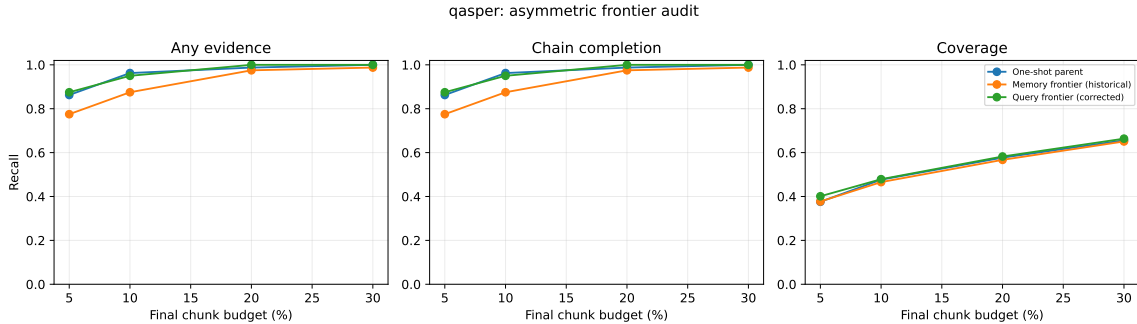


Figure 6: Gate-1 QASPER curves. Respecting the asymmetric projection contract recovers most of the historical closure loss, but one-shot remains the sparse-retrieval reference.

Projection asymmetry therefore explains a meaningful part of the original degradation. It does not explain why the corrected traversal fails to improve chain completion. That result passes the predeclared checkpoint for testing a finer propagation scale.

6.2 Gate 2: contextual local gists

Gate 2 changes the index to 256-token contextual parents with eight 32-token local gists. The one-shot and parent-closure baselines use the same parent features; local closure alone traverses

subchunk states. Parent deduplication holds final materialization to the same parent count.

The designed local-bridge control separates the methods cleanly. At two selected parents, one-shot and projection-correct parent closure retrieve only the entry parent: exact identity, chain completion, and coverage are 0, 0, and 0.5 over 320 examples. Local closure retrieves both evidence groups in every case, with exact identity, chain completion, and coverage all 1.0. Its mean Δ_{bridge} is 1.0. The implementation can therefore expose a bridge erased by a parent mean without re-encoding the local window.

Natural data is more restrained (Table 6). At 20%, local closure improves Hotpot any recall, exact identity, and coverage by 0.013 each over one-shot, but chain completion falls by 0.013 and parent closure is stronger. On QASPER, local closure improves exact identity by 0.025 and coverage by 0.016 over one-shot while any/chain recall falls by 0.013. Local accepted edges are stronger than corresponding parent edges: mean Δ_{bridge} is 0.203 on Hotpot and 0.205 on QASPER. That measurable locality does not yet identify the correct cross-parent chain.

Table 6: Gate 2 at a matched 20% final parent/KV budget. Comparisons count semantic gist dot products; no native QK comparisons occur in this gate.

Dataset	Method	Any	Chain	Exact	Coverage	Gist cmp.	K/V tokens
HotpotQA	One-shot parent	.475	.138	.000	.217	6.5	405.8
HotpotQA	Parent closure	.513	.175	.000	.235	11.9	405.0
HotpotQA	Local closure	.488	.125	.013	.229	49.5	403.4
QASPER	One-shot parent	.938	.938	.738	.813	18.2	1036.0
QASPER	Parent closure	.950	.950	.738	.819	66.6	1037.3
QASPER	Local closure	.925	.925	.763	.829	405.2	1031.0

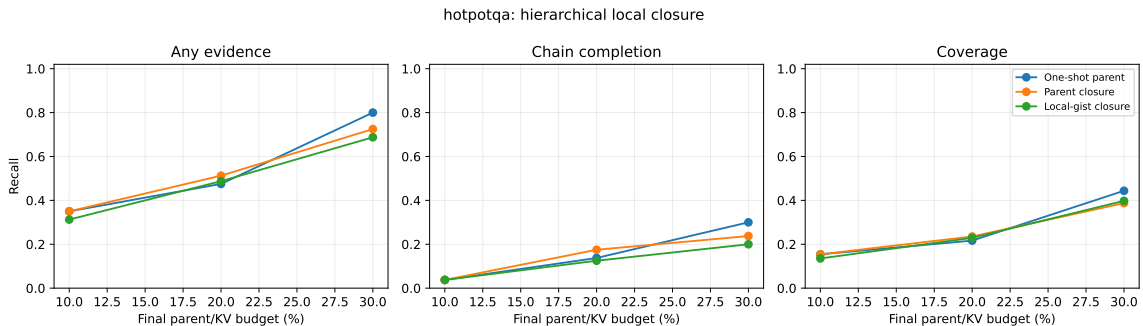


Figure 7: Gate-2 HotpotQA curves under matched final parent budgets. Local routing exposes stronger subchunk edges but does not improve evidence-group chain completion.

The cost is substantial. At 20%, local closure uses 49.5 versus 6.5 comparisons on Hotpot ($7.6\times$) and 405.2 versus 18.2 on QASPER ($22.3\times$). Materialized K/V differs by less than 1.3% within either dataset, confirming that the quality comparison is not purchased with a larger payload. The graph audit validates all 960 saved Gate-1/2 traces against schema v2 and finds no parent-budget violation.

6.3 Gate 3: native local QK propagation

The native-bridge control first verifies the mechanism without relying on natural annotations. The root-semantic rank retrieves entry parent A and a stronger semantic distractor, while a token in A has a model-compatible native QK match only to evidence parent B . Across five seeds and 64 cases per seed, one-shot, parent closure, and local semantic closure retrieve half the evidence and complete no chains. Native QK retrieves both groups in all 320 cases: exact identity, chain completion, and coverage are all 1.0. A distractor with an incompatible native key remains unselected. The vectorized scorer can therefore express the proposed transition.

Natural data does not reproduce that separation (Table 7). At the primary 20% final budget and 10% semantic candidate pool, max native QK matches local semantic chain completion on HotpotQA (0.125) and remains below one-shot (0.138) and parent closure (0.175). Its any-evidence recall and coverage are slightly higher than local semantic, but exact identity falls from 0.013 to zero. Expanding the pool to 20% lowers chain completion to 0.113. On QASPER, max native QK reaches 0.900 with the 10% pool and 0.863 with the 20% pool, below local semantic (0.925) and one-shot (0.938). More candidates amplify rather than repair propagation error.

Table 7: Gate 3 at a matched 20% final parent/KV budget. “QK-10” and “QK-20” are the primary max score with 10% and 20% semantic candidate-parent pools; “QK-Top4” is the sole Top-4 control at 20%. Native dots count token-pair/query-head products.

Dataset	Method	Chain	Coverage	Sem. cmp.	Native dots	K/V tok.	Time (s)
HotpotQA	One-shot	.138	.217	6.5	0	405.8	.0046
HotpotQA	Local semantic	.125	.217	98.3	0	407.0	.0105
HotpotQA	QK-10	.125	.223	114.6	95,712	406.3	.0121
HotpotQA	QK-20	.113	.208	114.6	183,667	407.2	.0132
HotpotQA	QK-Top4	.175	.265	114.6	183,667	406.0	.0143
QASPER	One-shot	.938	.813	18.2	0	1036.0	.0059
QASPER	Local semantic	.925	.829	540.0	0	1042.9	.0463
QASPER	QK-10	.900	.810	540.0	1,198,554	1043.7	.0467
QASPER	QK-20	.863	.794	540.0	1,949,958	1039.8	.0542
QASPER	QK-Top4	.838	.770	540.0	1,949,958	1041.3	.0624

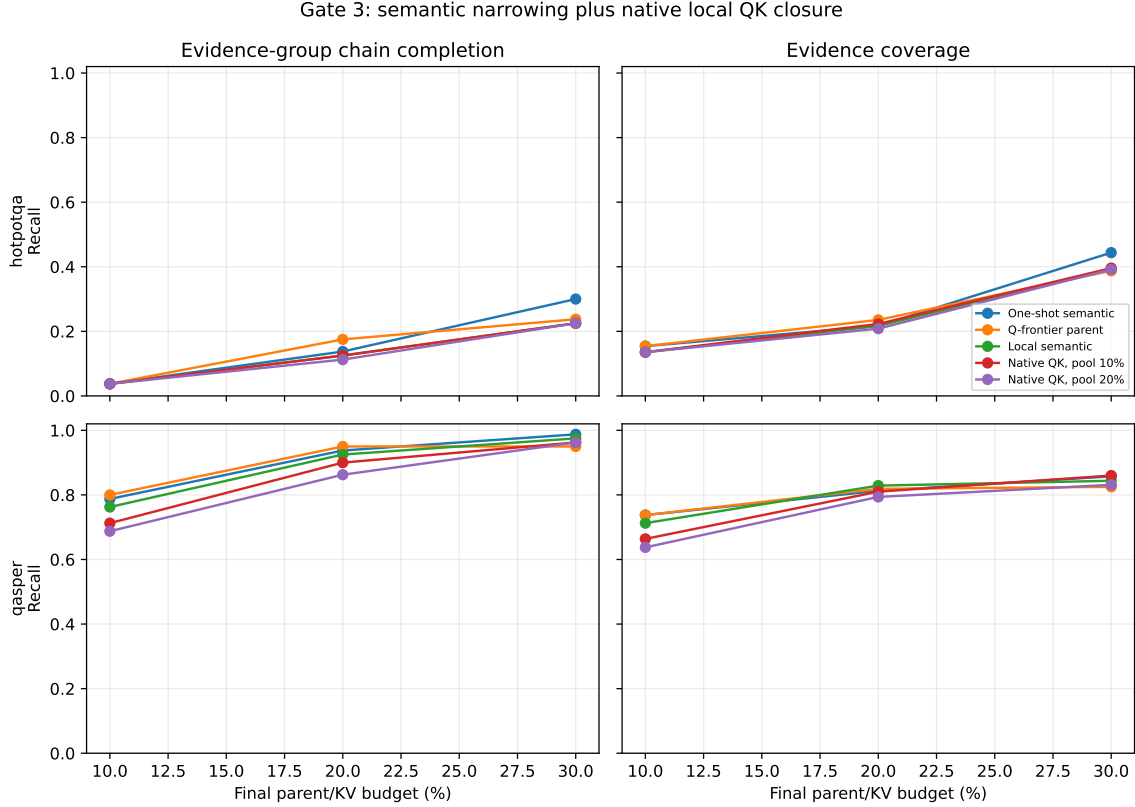


Figure 8: Gate-3 chain completion and evidence coverage across final budgets. Primary native-QK propagation does not dominate local semantic or one-shot routing; the larger candidate pool is usually worse, especially on the direct-retrieval QASPER control.

The Top-4 control is the one localized positive: HotpotQA chain completion at 20% rises to 0.175, matching parent closure and exceeding both one-shot and local semantic. It does not persist as a dominant curve: at 10% all methods tie at 0.038, at 30% Top-4 reaches 0.238 versus one-shot 0.300, and QASPER degrades to 0.838 at 20%. The $\mu + \sigma$ threshold is effectively identical to fixed Top- k in this hard-cap regime and offers no rescue. These predeclared controls are therefore a mixed point observation, not a positive Gate-3 result. This is specifically a failure of max-reduced Top-1 native closure under the tested competition rule; it does not yet show that useful native associative edges are absent. We test that distinction below.

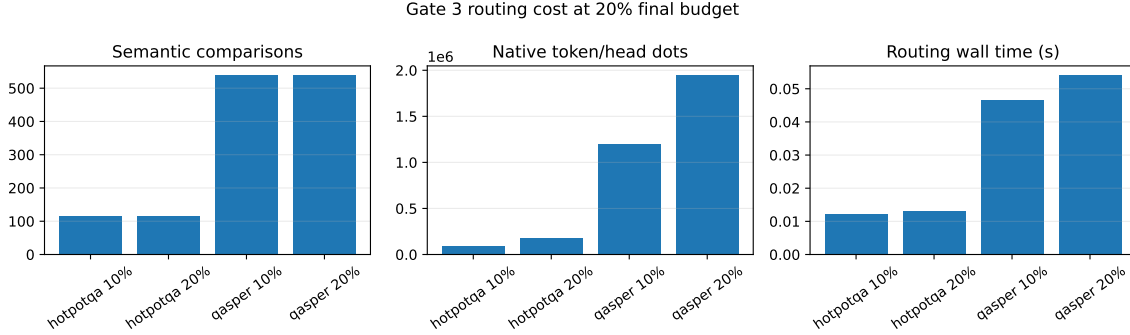


Figure 9: Gate-3 cost at the 20% final parent budget. Semantic and native work are reported separately. Candidate expansion from 10% to 20% nearly doubles native dot products without a quality gain. Wall time measures this batched reference implementation, not an optimized kernel.

At 20%, native refinement compares means of 1.1/1.7 candidate parents and 161/315 candidate tokens on HotpotQA for the 10/20% pools; QASPER compares 3.4/5.6 parents and 874/1,422 tokens. This produces 95,712–183,667 native dots on HotpotQA and 1.20–1.95 million on QASPER per example/seed, while materialized K/V stays within 0.8% of one-shot. Every saved graph reports zero K/V materializations during closure. Gate 3 changes search work and selected identities, not payload quantity or transformer attention semantics.

6.4 Oracle convergence, displacement, and competition

The previous gates score whether a selected set touches each evidence group. Here we ask the stricter question of whether it approaches the exact feasible parent set used by the annotated Paper-2 oracle. We import that oracle constructor unchanged, pass it the cached evidence spans and 256-token parent boundaries, and assert equality with the feature mask for every example. For selected set $S(B)$ and oracle S_O , we report recall, precision, Jaccard overlap, and $\mathbf{1}[S_O \subseteq S(B)]$ at 5, 10, 20, 30, and 40% parent/KV budgets. Figure 10 shows that Hotpot convergence remains slow for every frozen method. At 20%, parent closure has the best non-control recall (0.235); the native Top-4 reduction reaches 0.265, but complete recovery is only 0.013. At 40%, one-shot is strongest at 0.529 recall, while all iterative variants remain between 0.438 and 0.477. Among the few Hotpot seed/example trajectories that ever recover the complete oracle, the mean minimum budget is 31.7–34.0%; only 10–12 of 80 trajectories do so by 40%.

QASPER provides the expected direct-retrieval control. At 20%, one-shot reaches 0.813 oracle recall and 0.738 complete recovery. Parent and local closure reach 0.819/0.738 and 0.829/0.763, respectively, but native max falls to 0.794/0.725 and the Top-4 reduction to 0.770/0.713. At 40%, one-shot and local differ by only 0.002 recall. Iteration therefore does not buy a useful convergence advantage on this mostly one-group task.

That aggregate comparison hides a specific failure. At 20%, one-shot touches at least one oracle parent in 0.475 of Hotpot and 0.938 of QASPER seed/example runs. Of those successful roots, local, native-max, and native-Top-4 closure fail to preserve every touched oracle parent in 0.132/0.052, 0.184/0.117, and 0.158/0.156 of Hotpot/QASPER runs. Parent closure displaces none on Hotpot and 0.013 on QASPER. This is not literal eviction after admission: the frozen implementation reserves approximately half of B for root retrieval, keeps admitted roots, and fills the remaining slots through propagation. A “displaced” one-shot hit therefore falls outside the smaller iterative root allocation and is not recovered. Every replacement observed in this diagnostic is a hop-2

proposal.

Displacement is associated with root uncertainty rather than a larger final payload. For QASPER local closure, displaced cases have mean root Top-1–Top-2 margin 0.037 and entropy 1.842, versus 0.123 and 1.310 for preserved cases. Native-max cases show the same direction (0.089/1.728 versus 0.122/1.286), and native Top-4 cases remain weaker but consistent (0.104/1.618 versus 0.121/1.286). These are descriptive associations over only 4, 9, and 12 displaced runs, not a fitted confidence rule.

We isolate that mechanism with a matched- B protected-root counterfactual. It first preserves one-shot oracle hits, then fills the remaining slots in the frozen iterative order. This uses oracle labels and is an upper bound, not a deployable routing policy. On QASPER it raises oracle recall by 0.006 for parent closure, 0.015 for local closure, 0.044 for native max, and 0.068 for native Top-4; the two native methods both reach 0.838 recall and 0.763 complete recovery. Hotpot gains are smaller (0.000–0.031 recall) because its dominant error remains initial root discovery.

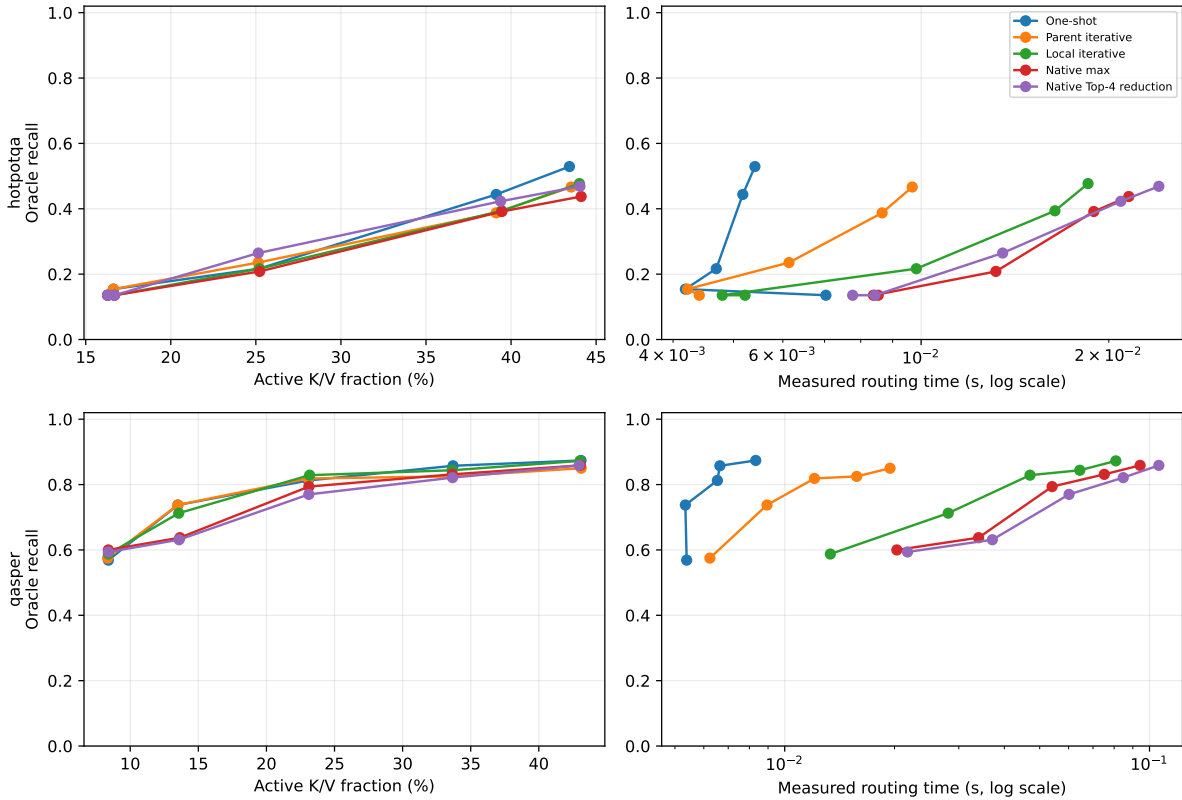


Figure 10: Exact Paper-2 oracle recall against active K/V fraction (left) and measured routing time (right), with HotpotQA above QASPER. The Top-4 label averages four token/head matches; it is not a four-parent beam. Iterative search adds routing work but supplies no cost-adjusted Hotpot advantage. Times describe the batched reference runner, not an optimized kernel.

Table 8: Oracle convergence and root preservation at the primary 20% parent/KV budget. “Displaced” is conditioned on a one-shot oracle hit. Protected recall is an oracle-informed counterfactual at the same B . Semantic comparisons and native dots are different units.

Dataset	Method	Recall	Complete	Displaced	Protected R	Sem. cmp.	Native dots
HotpotQA	One-shot	.217	.000	–	.217	6.5	0
HotpotQA	Parent iterative	.235	.000	.000	.235	11.9	0
HotpotQA	Local iterative	.217	.013	.132	.242	98.3	0
HotpotQA	Native max	.208	.000	.184	.240	114.6	183,667
HotpotQA	Native Top-4 reduction	.265	.013	.158	.290	114.6	183,667
QASPER	One-shot	.813	.738	–	.813	18.2	0
QASPER	Parent iterative	.819	.738	.013	.825	66.6	0
QASPER	Local iterative	.829	.763	.052	.844	540.0	0
QASPER	Native max	.794	.725	.117	.838	540.0	1,949,958
QASPER	Native Top-4 reduction	.770	.713	.156	.838	540.0	1,949,958

Low end-to-end recovery can arise before or after the first evidence group. At 20%, one-shot reaches the first source-ordered Hotpot group in 0.200 of seed/example runs; parent closure reaches 0.213, local closure 0.200, native max 0.188, and native Top-4 0.263. To isolate propagation, we then force only that annotated source group active and rank the next evidence group among all other parents. Any local node in either evidence parent may realize an edge. Parent semantic uses the projection-correct query-to-memory score, local semantic uses the strongest local pair, and native QK uses the exact frozen pre-RoPE max token/head reduction.

The conditioned result is much stronger than end-to-end closure (Figure 11). Median true-edge rank is 3 in all three spaces. Parent semantic, local semantic, and native QK respectively obtain R@1 of 0.141, 0.259, and 0.235; R@4 of 0.765, 0.741, and 0.765; and MRR of 0.424, 0.480, and 0.469. Every semantic transition is within Top-8, as are 16 of 17 native transitions. Mean oracle-minus-best-distractor margins remain negative ($-0.118, -0.143, -0.039$), so the correct edge is usually present but loses hard competition. Across 85 seed-transition comparisons, the best semantic and native ranks are both within Top-4 in 56 cases; semantic alone is near-top in 14, native alone in 9, and both are poor in only 6.

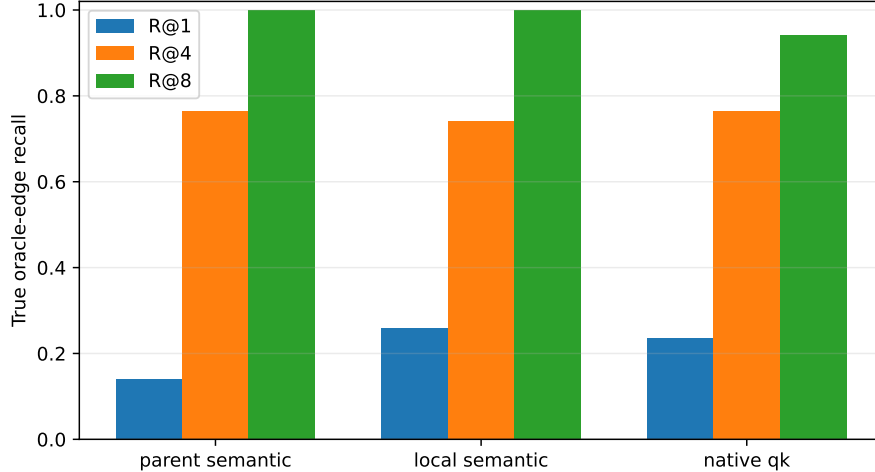


Figure 11: Rank recall for the true next Hotpot evidence group after conditioning on the first group. Most annotated transitions exist within the first four candidates even though they rarely win Top-1. Semantic statistics contain 17 transitions under five projection seeds; native QK has 17 seed-invariant transitions.

The Gate-3 Top-4 signal should not be read as parent-beam R@4. It averages four strong token/head matches before ranking parent transitions. Its four 20% chain wins over max reduction comprise two questions. In one seed-case a near-tied oracle edge moves from rank 2 to rank 1: its margin changes from -0.0038 to $+0.0157$. The other question accounts for three seed-cases, but its source-ordered edge remains rank 6 under both reductions; completion comes from root/reverse-order selection rather than better forward propagation. The cached features contain no entity/topic labels, so we do not assign a semantic relation to those distractors from parent IDs alone.

We also audit whether incomparable score families create the competition error. They never enter a common raw TopK in the frozen router: root and semantic propagation use normalized cosines followed by a 0.25 root anchor and $[0, 1]$ affinity/path products, while native QK uses a sigmoid score followed by its own 0.25 anchor and path product. The audit nevertheless reveals severe native compression. At 20%, maximum and Top-4 native logits average 13.95/12.16 on Hotpot and 14.29/12.47 on QASPER; every sigmoid exceeds 0.999, and 94.8–100% exceed 0.99999. Top-4 does improve the Hotpot oracle-edge score quantile from 0.561 to 0.731, consistent with less sensitivity to one extreme token/head match, but it does not generalize to QASPER.

As an offline control, we fix the candidate identity union already exposed by one-shot and iterative routing, fit per-family z-score or empirical-quantile transforms on a hash-stable validation partition, and evaluate unchanged on held-out examples. The transforms cannot discover a missing parent. They leave QASPER local recall at 0.900 and raise native-max recall from 0.867 to 0.883; quantiles raise native Top-4 from 0.833 to 0.883. They do not improve Hotpot recall: the best normalized native result remains 0.205, and local z-scoring remains 0.190. Calibration can preserve a strong root once it enters the candidate pool, but does not solve Hotpot’s missing-root problem.

Finally, we simulate, without changing SDK routing, $k = 1$ when the Top-1/Top-2 gap exceeds a validation-selected threshold and $k = 4$ otherwise. On held-out edges, parent semantic selects $k = 4$ always and reaches 0.686; local semantic averages $k = 3.83$ and reaches 0.657 versus its 0.686 fixed-R@4 ceiling; native averages $k = 3.57$ and reaches 0.714, exactly its fixed R@4. The split is only 10/7 distinct native transitions and 50/35 seed-sensitive semantic transitions, so this is a mechanism diagnostic, not a tuned policy result. The simple margin rule saves little competition

and does not explain the Top-4 reduction signal.

These diagnostics motivate the deployable, label-free competition gate evaluated next. A production policy cannot know oracle roots, but it can lock identities using root confidence or cross-seed agreement, then let propagation compete only for remaining slots. This directly tests whether the protected-root upper bound and near-Top-4 Hotpot edges survive without oracle access.

6.5 Monotonic root persistence and adaptive competition

The previous router divides B before seeing the score distribution. We instead compute the ordinary full root ranking $R = (r_1, \dots, r_B)$, choose a label-free lock set $L \subseteq R$, and enforce

$$L \subseteq S_{\text{final}}, \quad B_{\text{prop}} = B - |L|, \quad |S_{\text{final}}| = B. \quad (14)$$

Propagation can compete only for the B_{prop} unlocked slots. Duplicate parent proposals collapse to their strongest path; if propagation supplies too few unique identities, the original root order fills the remainder. The equality above therefore changes identities, not active parent/KV capacity. No evidence label appears in the policy API.

We test fixed prefixes of 1, 2, and 4, first-large-score-drop thresholds, full-vector root z -score thresholds, and the fraction of five router seeds placing a parent within root Top- B . For transitions we retain projection-correct local semantics and the frozen Top-4-mean pre-RoPE native-QK scorer. Native magnitudes are saturated after a sigmoid, so cross-source native proposals use deterministic per-source raw-logit rank. Adaptive breadth chooses $k = 1$ for concentrated semantic/native Top-1 agreement, $k = 2$ for moderate concentration with Top-4 overlap, and $k = 4$ otherwise. Thresholds and policy identities are selected on the hash-stable validation examples only. We freeze two roles per geometry: the unconstrained preservation winner and the best exploratory root policy required to activate propagation on at least 25% of validation runs. This second role prevents a one-shot-equivalent winner from making transition competition vacuous.

Validation selects root $z > -0.5$ for preservation in both geometries. On this sample that rule locks all available Top- B identities, performs no propagation, and becomes an independently implemented one-shot parity check. For exploration, validation selects root $z > 1$ with native rank and seed agreement at 0.6 with semantic propagation. Fixed $k = 1$ wins for native rank; fixed $k = 4$ wins for semantic propagation. None of the nine adaptive threshold pairs wins. Across held-out adaptive-policy traces, native decisions are mostly $k = 1/2$ (1,674/1,014 source decisions, with 30 at $k = 4$); the richer rule does not expose a natural-data gain hidden by fixed breadth.

Table 9: Held-out results at the primary 20% final parent/KV budget. Seven HotpotQA and six QASPER examples are each evaluated under five frozen router seeds. “Prop.” is the fraction of seed/example runs that add a propagated identity. The preservation rows lock full root Top- B and therefore exactly match one-shot.

Dataset	Policy	Oracle R	Chain	Root present	Root locked	Prop.	Native dots
HotpotQA	One-shot	.205	.143	.343	.343	.000	0
HotpotQA	Monotonic preserve	.205	.143	.343	.343	.000	0
HotpotQA	Native-rank explore	.233	.143	.343	.314	.314	82,388
HotpotQA	Semantic explore	.176	.086	.343	.257	.314	82,388
QASPER	One-shot	.883	.800	.967	.967	.000	0
QASPER	Monotonic preserve	.883	.800	.967	.967	.000	0
QASPER	Native-rank explore	.883	.800	.967	.967	.767	907,281
QASPER	Semantic explore	.883	.800	.967	.967	.400	621,193

The result separates preservation from discovery. Monotonicity removes the measured QASPER regression exactly, but the selected exploratory policies spend search work without improving 20% chain completion. Native rank gains 0.028 Hotpot oracle recall (0.205 to 0.233) while chain completion remains 0.143. At 30/40%, native chain completion falls from one-shot 0.343/0.371 to 0.257/0.286. Semantic exploration matches one-shot chain completion at 30/40% (0.343/0.371), and reaches QASPER 0.950 recall and 0.900 chain completion at 40%, but it does not produce the required Hotpot advantage. The dual-geometry diagnostic scorer takes about 0.006 seconds per Hotpot and 0.044–0.068 seconds per QASPER run at 20%; these reference timings include the alternate geometry used for confidence measurement and are not an optimized kernel claim.

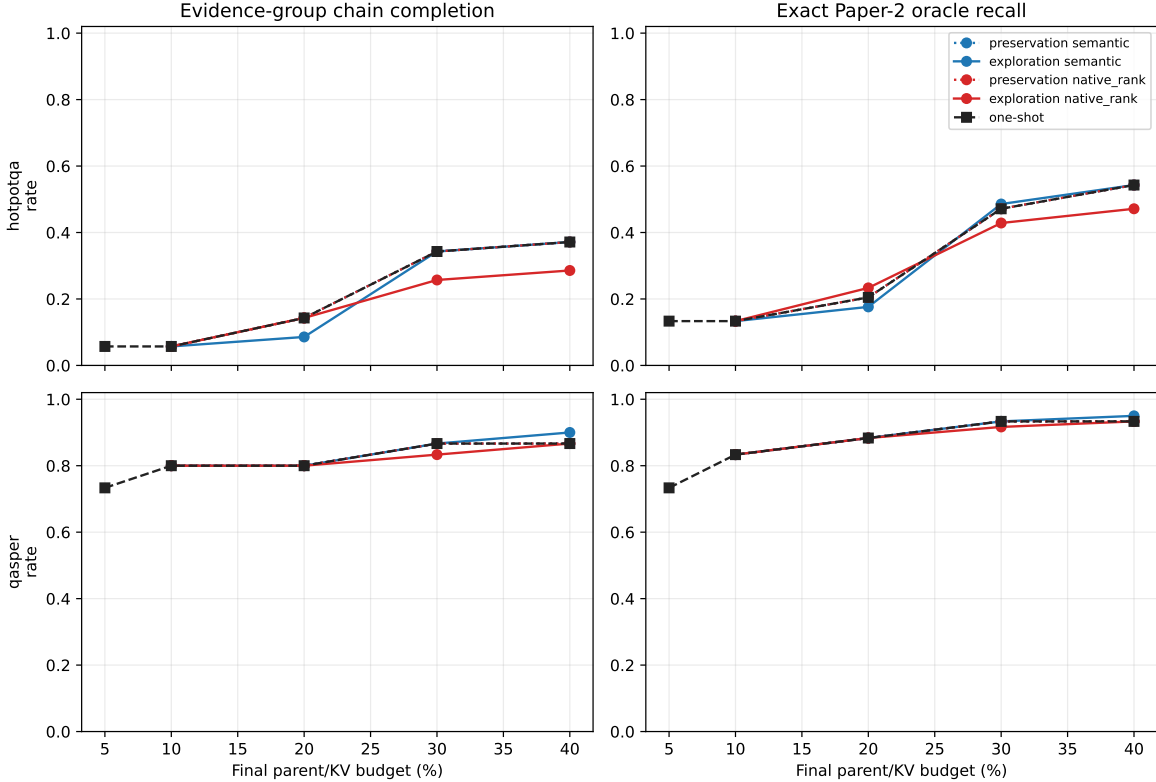


Figure 12: Held-out quality under matched final budgets. Dotted preservation curves reproduce one-shot. Exploratory propagation does not improve Hotpot chain completion; the isolated 20% native oracle-recall gain does not persist as budgets grow.

The failure decomposition is more decisive than the aggregate delta. Across exploratory held-out runs and budgets, the first source-ordered Hotpot evidence group is absent from full root Top- B in 52.9% of cases. It is present but rejected by the lock in 11.1%; the remainder is locked. At 20%, the native policy locks the first group in 31.4% of runs and recovers all later groups in 45.5% of those conditioned runs, yet end-to-end chain completion still cannot exceed one-shot. Competition policy cannot propagate from an entry that was never admitted.

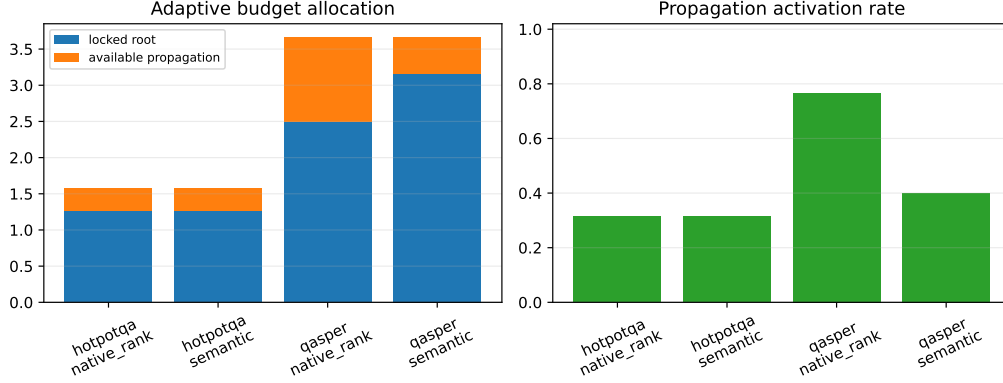


Figure 13: Held-out exploratory budget allocation and propagation activation at 20%. The final parent budget remains matched even when the locally selected lock count differs.

Controlled cases verify the mechanism independently of that natural limitation. Obvious direct retrieval locks the root and stops. With the target outside root Top- B , fixed $k = 1$ obtains 0.5 oracle recall on both an ambiguous rank-2 transition and a distractor-heavy rank-4 transition. Fixed $k = 4$ and adaptive 1/2/4 both recover 1.0 at the same final B ; the adaptive rule chooses 2 and 4, respectively. Thus persistence and adaptive breadth are implemented and can regulate designed cases, but validation does not promote them on natural data.

This gate therefore advances a narrower recommendation: refine entry discovery before changing transition geometry. The next experiment below tests whether the pretrained contextual question already contains a useful entry relation that the single final-token query collapses.

6.6 Contextual query facets and native-head entry search

The inherited root is one attention-input hidden state, $h_q = H_q[T - 1]$, projected by the frozen query map and cosine-scored against each fixed parent memory gist. We reproduce this operation exactly as condition A. For condition B, the model still sees the complete prompt exactly once. Within the recorded question span, overlapping windows produce

$$g_{q,i} = \frac{1}{b_i - a_i} \sum_{t=a_i}^{b_i-1} H_q[t], \quad s_j = \max_{i \in \{\text{global}, \text{local}\}} \cos(W_q g_{q,i}, W_m h_j). \quad (15)$$

The global final-token row remains in the set. All facets nominate through one score per parent and one final Top- B ; no facet receives a private payload budget. We test 4/8/16-token windows at half-window stride and max versus Top-2-mean facet reduction. Validation selects width 4, stride 2, and max. The query-feature manifest verifies one full prompt forward per example, zero independent window forwards, and bit-exact equality between condition A and the frozen root cache.

The learned 128-D semantic router has no legitimate head axis. We do not invent pseudoheads. Conditions C and D instead audit the model’s 16 real layer-27 pre-RoPE query heads against their 8 grouped-query-compatible K heads. Valid local K tokens are vectorized into a parent mean; each query head h compares only with K/V head $\lfloor h/2 \rfloor$. Validation selects independent per-head Top-1 nominations followed by deduplication and one global Top- B rerank. Because this changes semantic projections to native QK, C/D are an operational head audit, not a strict factorial contrast with A/B. Native mean-head controls E/F isolate span effects within that representation.

Table 10: Held-out root-entry quality at the primary 20% parent budget. A/B average seven HotpotQA or six QASPER identities over five frozen router seeds. C/D are deterministic native-QK conditions and are not replicated as fictitious seeds. “Entry” means that the first ordered evidence group occurs in root Top-B.

Dataset	Variant	Entry	Rank	MRR	R@1	R@2	R@4	R@8
HotpotQA	A: global semantic	.343	2.60	.503	.229	.486	.971	1.00
HotpotQA	B: multi-span semantic	.457	2.23	.626	.429	.571	.943	1.00
HotpotQA	C: global split-head	.429	3.00	.585	.429	.571	.714	1.00
HotpotQA	D: multi-span split-head	.143	3.57	.379	.143	.286	.714	1.00
QASPER	A: global semantic	.967	1.60	.811	.700	.800	.967	1.00
QASPER	B: multi-span semantic	.967	1.50	.828	.700	.900	.967	1.00
QASPER	C: global split-head	.667	4.67	.292	.000	.333	.500	.833
QASPER	D: multi-span split-head	.667	5.50	.232	.000	.167	.333	.833

At 20%, multi-span semantic search improves Hotpot entry presence by 0.114, oracle recall from 0.205 to 0.295, mean rank from 2.60 to 2.23, and oracle precision from 0.343 to 0.457. Its mean oracle margin remains negative but narrows from -0.029 to -0.009 ; Jaccard rises from 0.167 to 0.248 and false-positive parents fall from 1.114 to 0.943. The paired entry changes are six gains, two losses, and 27 ties. At 10/30/40%, entry presence changes from 0.229/0.629/0.686 to 0.429/0.714/0.800. QASPER entry presence, oracle recall, precision, and Jaccard remain exactly 0.967, 0.883, 0.296, and 0.288; MRR rises slightly.

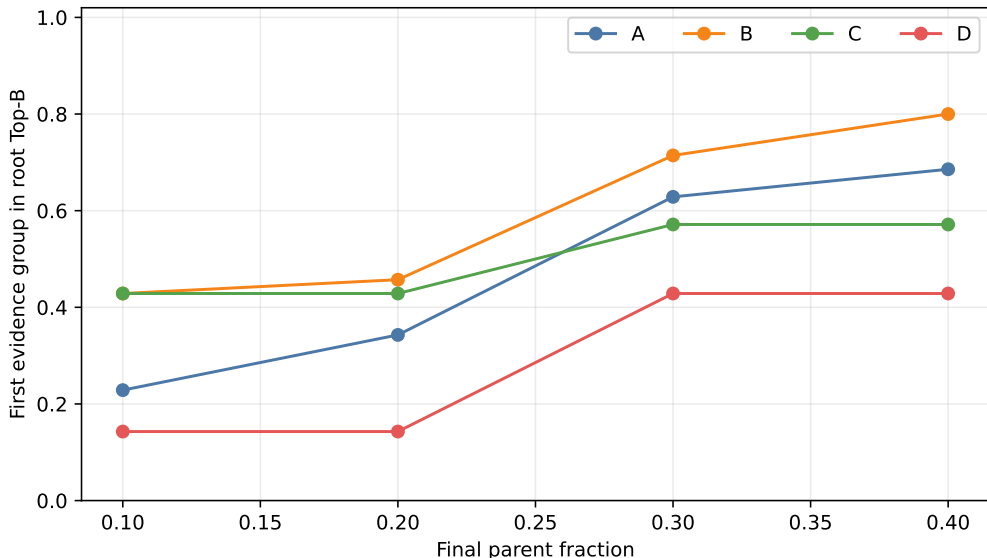


Figure 14: Held-out HotpotQA first-evidence entry across matched parent budgets. Contextual multi-span semantic facets (B) improve the exact global semantic control (A) at every budget. Native head splitting (C/D) does not combine with that gain.

The gain is not free search. Mean Hotpot query-parent comparisons rise from 5.9 in A to 67.6 in B, while C and D require 93.7 and 1,081.1 raw span/head comparisons. QASPER requires 16.3, 82.0, 261.3, and 1,312.0, respectively. Thus B gains about 0.00185 entry probability per extra Hotpot comparison. Batched GPU wall time is approximately 0.0048 seconds for both A and B

at this small scale; C/D take 0.017/0.016 seconds on Hotpot and 0.040/0.042 on QASPER. These are diagnostic timings, not optimized-kernel throughput. Native mean-head E/F controls move Hotpot entry from 0.286 to 0.429 but remain far below the semantic QASPER control (0.333 to 0.500 versus 0.967). The native representation change, not merely head aggregation, is therefore consequential.

The provenance audit finds six Hotpot seed/example cases in which A misses and B succeeds. Winning windows include relation-bearing fragments such as “Most Outstanding Defensive Player” and entity fragments containing “Scott”. Two paired losses and the still-negative mean margin warn against treating every local maximum as reliable. Four global-head specialization cases exist, but gains are concentrated and do not survive the native split-head aggregate. The span-head interaction audit finds no unique span-only or synergistic win in the deterministic native conditions. Separate span-dilution, head-specialization, and combined synthetic controls all recover their designed parent while conserving the final budget.

The predeclared 0.10 Hotpot gain gate is met without QASPER loss, so we run the small confirmation in Table 11. It freezes B and reconnects only the previously selected $z > 1$ root lock, native-rank transition geometry, and fixed Top-1 transition. No policy is reselected on held-out labels.

Table 11: Held-out confirmation at the same 20% final parent/KV budget. “Exact” requires every canonical oracle parent; “Chain” requires one parent from every ordered evidence group.

Method	HotpotQA				QASPER		
	Entry	Oracle R	Exact	Chain	Entry	Oracle R	Chain
Current one-shot	.343	.205	.000	.143	.967	.883	.800
Current monotonic iterative	.343	.233	.000	.143	.967	.883	.800
New faceted root only	.457	.295	.029	.171	.967	.883	.800
Faceted root + propagation	.457	.295	.057	.171	.967	.883	.800

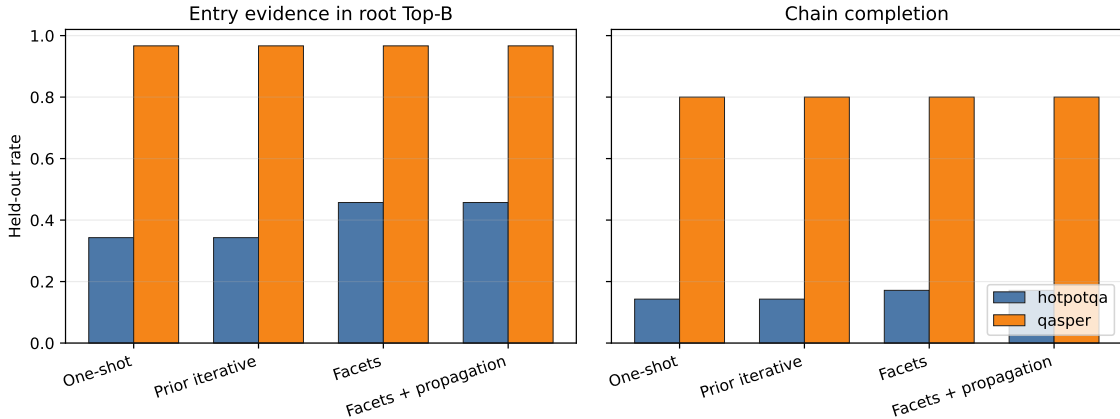


Figure 15: Entry discovery and downstream chain completion for the four confirmation methods. Facets improve Hotpot entry and chain metrics while preserving QASPER; propagation does not add chain completion beyond the new root at this budget.

The root-only facet search averages 68 comparisons on Hotpot and 82 on QASPER. Reconnecting the dual-geometry transition diagnostic raises these to about 112,232 and 1,154,036, respectively,

while leaving chain completion unchanged from B. Exact-oracle completion does rise by 0.029 on Hotpot, but the incremental search is not justified as a default. The evidence supports recommendation A: *use multi-span query facets* as an opt-in research selector. Query-span dilution is the dominant tested entry failure; head dilution and span-head synergy are not supported. One-shot remains the SDK default, and learned slots, textual decomposition, new associative training, and partial materialization remain outside this gate.

7 Discussion

7.1 What the experiments establish

The synthetic result and the integration tests establish three mechanical facts. First, a gist can serve as a new query and recover a node that the root cannot afford at matched *B*. Second, deduplication, beam pruning, and hard budgets prevent exponential materialization. Third, the closure can remain entirely in compact routing state until final identities map to full native K/V.

The natural-data results establish a different boundary. Projection-correct closure confirms that shared dimensionality did not imply shared query and memory semantics: respecting the asymmetric interface repairs much of QASPER’s loss. The local experiment then finds positive bridge deltas and better exact identity/coverage, so parent aggregation does hide some useful semantic relations. Yet those stronger local edges do not improve chain completion. Direct iterative reuse of the inherited one-shot routing representation does not yet provide a dependable associative traversal graph on natural data.

Gate 3 then tests the transformer’s own position-neutral local matching geometry after semantic narrowing. Its exact synthetic recovery proves that the routing path, grouped heads, and tokenwise reduction can represent a native bridge. The natural result is nevertheless negative under the predeclared success criterion: primary native QK does not beat local semantic closure or one-shot across matched budgets. The isolated HotpotQA Top-4 gain is offset by other budgets and by QASPER degradation. Initial relevance geometry and propagation geometry may differ, but pretrained layer-27 QK does not establish that specialization here.

This completes the staged diagnostic sequence: Stage 0 is one-shot retrieval; Stage 1 is chunk iterative closure; Stage 2 corrects the asymmetric projection interface; Stage 3 moves propagation to contextual local semantic gists; Stage 4 tests tokenwise native QK after semantic narrowing; Stage 5 makes root identity persistence monotonic and allocates the remaining budget through validation-selected bounded competition. Stage 6 preserves several contextual views of the question through root competition instead of forcing one relational summary before search. Projection role, propagation scale, and native attention geometry have therefore been tested before introducing learned associative supervision. Oracle-conditioned rank now shows that most true transitions survive within Top-4, so a learned head is not yet the next controlled intervention. The displacement audit separates two failures that the aggregate metric had mixed: QASPER often starts from the right parent and can lose it to the fixed root/propagation slot split, whereas Hotpot usually fails to discover the first group. Stage 5 confirms the distinction: monotonic persistence reproduces QASPER one-shot quality, but bounded competition cannot propagate from a missing Hotpot entry. Stage 6 then recovers part of that missing-root gap with fixed, parameter-free contextual facets. Its success and the failure of native split-head composition localize the tested problem to semantic query-span dilution. Propagation still does not add chain completion beyond the improved root, so any later learned query decomposition or associative component remains new machinery rather than an automatic extension.

7.2 Relation to retrieval, agents, and recursive models

Memory Networks repeatedly read differentiable memory to support multi-hop tasks, while RAG, RETRO, and interleaved retrieval systems bring external evidence into generation [4, 1, 10]. Agentic systems can reformulate a text query after each model pass. Iterative PRA is narrower: it traverses already computed latent addresses inside one memory subsystem. It avoids repeated text serialization and full-model round trips, but it cannot replace an agent that must plan tools, verify evidence, or change the retrieval language.

Tiny Recursive Models repeatedly transform internal latent and answer state [3]. Controlled autoregressive work finds no reliable gain from the full autoregressive TRM architecture over simpler refinement baselines [6]. Iterative PRA does not apply a tiny neural network repeatedly. Its recurrence is *query* $\rightarrow$ *memory node* $\rightarrow$ *memory node*. This distinction motivates the parameter-free Level-1 test: retrieval recursion should earn its complexity before neural state recursion is introduced.

7.3 Paper 3 handoff

Papers 1–2 conservatively compress search but materialize every selected chunk in full. Iterative closure produces more structure before transfer: hop, parent edges, direct relevance, path score, confidence, branch entropy, and duplicate/overlap statistics. The versioned graph keeps these signals even when the final payload remains unchanged. Paper 3 can therefore ask a separate question: given a discovered closure, how much detail should each node receive? This paper does not test partial materialization.

8 Limitations

The natural-data pool has 16 examples per dataset, repeated over five router seeds and then split by stable example identity for the final policy-selection gate. Seeds quantify projection sensitivity but do not create 80 independent questions. Hotpot evidence groups are source-contiguous spans, not annotated causal hop labels. QASPER’s selected subset is yes/no and often one-group, so it tests direct retrieval more than multi-hop closure.

The protected-root result is an oracle-labeled upper bound, not an implementable policy. The calibration controls rerank only the fixed union of identities already exposed by frozen routing; they cannot recover an unseen candidate and should not be read as a trained-router comparison. Their validation/test partitions are example-disjoint, but both remain small.

The adaptive-competition policy family is selected on nine HotpotQA and ten QASPER validation examples and reported on seven and six held-out examples, respectively; five router seeds measure projection sensitivity but do not enlarge those example counts. The validation objective weights Hotpot chain/recall, QASPER recall, and QASPER propagation cost explicitly. Different task weights could select another transparent rule, although the full selection audit is retained. The preservation winner degenerates to full Top-*B* locking on this sample and should be interpreted as a parity result, not evidence that $z > -0.5$ generalizes as a universal threshold.

The natural-data sample remains small. Projection correction, local hierarchy, native QK, and contextual query facets remove four specific confounds, but the experiment does not include adaptive zoom rules, evidence-edge training, a layer sweep, or a full agentic baseline. The local and native bridge controls are deliberately geometric and do not approximate natural language. Hotpot evidence groups are approximate source-order targets rather than annotated transition edges, so they can understate or misidentify the useful successor relation. Downstream F1/EM is measured

only in a two-example execution smoke and is not used as evidence of quality. Finally, the scorer batches token pairs and heads but still launches small per-example routing operations; the reported wall time is not a production-kernel claim.

The facet selector is chosen on nine HotpotQA and ten QASPER validation identities and evaluated on only seven and six held-out identities. Five frozen projections expose sensitivity but are not independent questions. Max over more facets creates a multiple-comparisons effect even under one final budget; margins, false positives, precision, and Jaccard are reported, but a larger corpus is needed to estimate calibration. The native-head conditions necessarily change from the learned semantic representation to pre-RoPE QK because the semantic router has no head axis; they cannot identify a pure head effect relative to A/B. Finally, the 4-token winning windows can be grammatical fragments rather than explicit subquestions. The result demonstrates useful contextual localization, not interpretable decomposition or causal hop annotation.

9 Conclusion

Bounded associative closure is exact when latent memory contains the right edges. Correcting the asymmetric frontier is necessary and materially reduces inherited degradation. Routing at a finer local scale can recover bridges erased by parent means. Tokenwise pre-RoPE native QK then recovers every designed native bridge, showing that semantic narrowing and a different propagation primitive can compose without changing materialization. None of these parameter-free *propagation* variants produces a dependable natural-data chain advantage at matched payload budget, and each finer search adds work. One-shot PRA therefore remains the default efficiency/quality reference even after the separate query-facet gain reported below.

The scientific lesson is narrower than “pretrained geometry has no associative edges.” Direct iterative reuse is sensitive to projection interface, propagation scale, candidate width, and score reduction. Local semantic and native attention spaces contain designed bridges and occasional natural positives, but neither supplies reliable cross-parent transitions on this sample. The oracle-rank diagnosis narrows the next step further: useful transitions are commonly present below Top-1, while root evidence discovery remains weak. The final frozen-policy gate confirms both sides. Monotonic identity locks preserve direct QASPER evidence exactly, and adaptive breadth recovers controlled rank-2/rank-4 edges, but neither yields a held-out Hotpot chain advantage. In 52.9% of exploratory Hotpot runs the first evidence group is absent before locking begins. The next controlled question is therefore entry discovery through local query facets, not another transition geometry or a larger breadth sweep. Fixed contextual facets do admit more missing Hotpot roots: at 20%, entry presence reaches 0.457 from 0.343 and oracle recall reaches 0.295 from 0.205, while QASPER recall remains 0.883. The resulting Hotpot chain rate is 0.171 versus 0.143, but reconnecting bounded propagation adds no further chain gain and costs orders of magnitude more comparisons. Multi-span entry search is therefore the supported opt-in extension; one-shot remains the default. Any learned extension must retain the same payload, split, leakage, and execution controls and improve beyond this stronger faceted root.

A Implementation Map

The implementation is designed so another system can reproduce the mechanism without adopting the entire PRA-HF package. File and line references below refer to the accompanying source revision.

Layer-local packed index. `src/prahf/iterative.py:69--134` packs variable gist sets as `[chunks,max_gists,routing_width]` plus a validity mask. Records retain chunk identities and payload handles, but packing copies no native K/V.

Asymmetric companions and contextual parents. Lines 387–438 of `injection.py` project each contextual local gist with both W_m and W_q and projects the occupancy-weighted parent hidden-state mean separately. The full path is `src/pratorch/hf/injection.py`. The cache stores compact companion vectors, not another native K/V payload.

Hierarchical index and parent deduplication. Lines 137–254 of `iterative.py` define aligned parent/local tensors and construct them from contextual segment gists. Lines 641–787 perform root-to-parent/local and local-to-local expansion, aggregates candidates by parent, enforces the unique-parent budget, and maps only final parents to payload handles.

Graph contract. Lines 269–342 of `iterative.py` define node, edge, graph, and result records. The current JSON Schema is `docs/papers/shared/schemas/retrieval_graph_v2.schema.json`. Nodes retain parent identity, local span, resolution, representation and projection types, hop, path score, and materialization state. Native-edge extensions additionally retain source/target spans, query and K/V head identities, native score, threshold, and semantic candidate rank.

Exact scoring and deterministic top-k. `iterative.py:376--397` normalizes frontier rows and computes `einsum("fd,cgd->fcg")` before max reduction across gists. A stable index tie-break is applied before `torch.topk`, producing deterministic identities on CPU and CUDA.

Traversal. `iterative.py:399--614` performs root scoring, frontier expansion, anchoring, deduplication, path pruning, Level-2 updates, stopping, and cost accounting. The code never accesses `chunk.token_kv`. Conversion at lines 617–638 returns selected payload handles only after the graph is final.

Host-model integration. `src/prahf/model.py:374--439` captures one root query with memory disabled, dispatches one-shot, chunk iterative, or opt-in `local_iterative` routing, maps final parent identities to consumption layers, enables fixed memory, and only then marks graph nodes materialized. Generation thereafter uses the unchanged Paper 2 native-K/V path.

Native QK feature capture. `precompute_native_qk_features.py:50--107` encodes each bounded 256-token parent once, reads Qwen’s exact layer-27 capture, transposes canonical Q/K to token-major form, and slices aligned 32-token regions. It stores FP16 pre-RoPE Q/K with a validity mask for a short final window; it never independently re-encodes a local region. The capture itself is generated by Qwen’s ordinary `q_proj/k_proj` and native Q/K norms before rotary position encoding. A test recomputes those projections from the captured hidden states and requires exact tensor equality.

GQA-compatible tokenwise scoring. Lines 122–211 of `src/prahf/native_closure.py` map query head h to K/V head $\lfloor h/2 \rfloor$, packs candidate keys accordingly, and computes all frontier-by-candidate token/head products with one `einsum`. Invalid padded pairs become $-\infty$ before max or Top-4 reduction. The scorer returns strongest token and compatible-head identities as well as the physical dot-product count; no Python loop traverses token pairs or heads.

Semantic narrowing and native propagation. `native_closure.py:215--480` uses root semantic scores for initial parents, applies $W_q h_A \rightarrow W_m h_B$ to rank a hard 10/20% candidate-parent pool, gathers all contextual local windows in that pool, and invokes native QK once. Proposals are merged by parent before the hard final parent budget. Threshold mode retains the same cap and fills rejected slots from root semantic ranking, preserving materialization parity. The graph cost record separates semantic comparisons, native dots, candidate parents/regions/tokens, accepted transitions, selected parents, final K/V tokens, and routing wall time.

Oracle and edge-rank diagnostic. Lines 66–116 of `run_oracle_convergence.py` map each annotation to overlapping parents and call the Paper-2 oracle constructor unchanged; disagreement with the cached evidence mask is fatal. Lines 133–185 define target rank using only frozen score vectors, with equal scores sharing a rank and the forced source group excluded. Parent/local/native score vectors are constructed at lines 188–245; target identity enters only the subsequent metric. Lines 252–292 fit the offline margin threshold on a hash-stable validation partition and evaluate it unchanged on test. The main loop beginning at line 588 wraps the existing Gate-2/3 evaluators, so convergence does not duplicate or subtly alter their selection policy.

Displacement and calibration diagnostic. Lines 81–159 of the additive displacement runner listed in the reproduction README compare one-shot oracle hits with the final identities returned by each frozen evaluator and retain replacement hop, score family, rank, and budget reason. Lines 162–174 implement the oracle-labeled protected-root counterfactual with the same final B ; this function is used only by the offline runner. Lines 184–277 construct a fixed candidate-identity union, fit score-family statistics on validation examples only, and rerank held-out examples by raw, z-scored, or empirical-quantile score. The root/semantic and native score capture at lines 337–478 follows the exact transformations used by Gate 2 and Gate 3. The main loop at lines 696–1095 calls those frozen evaluators, verifies all 4,000 selections row-for-row against the prior convergence artifact, and writes additive rows without changing SDK routing.

Fixed-identity payload parity. `tests/test_hf_integration.py` sends the same selected chunk through router, oracle, and native-closure labels. All three paths produce byte-identical post-RoPE K, native V, and attention output. Gate 3 therefore ends at identities: it neither changes nor reconstructs the payload consumed by Qwen.

Monotonic adaptive competition. `src/prax_hf/adaptive_competition.py:44--154` defines immutable root/transition policies and score/result records without any oracle field. Lines 157–234 compute cross-seed agreement and choose a lock only after constructing full root Top- B . Lines 237–291 standardize raw pre-RoPE native scores per source, measure Top-1/Top-4 spread, normalized entropy and semantic/native rank agreement, then select bounded breadth 1, 2, or 4. Lines 294–323 merge parent proposals while asserting that every locked identity survives; root order fills any unused slots. The opt-in router at lines 326–404 invokes frozen transition providers only when $B - |L| > 0$, uses raw native rank rather than sigmoid magnitude, and returns separate root, semantic, native-dot, and scorer time costs. It owns no K/V tensor.

The additive runner constructs projection-correct semantic and native traces at lines 162–247 of `run_monotonic_adaptive_competition.py`. Lines 285–383 call the label-free router, then attach exact-oracle and evidence-group metrics to its returned identities. Validation-only selection is at lines 475–509; synthetic direct/rank-2/rank-4 controls and cost-normalized held-out effects are at lines 551–692. The main loop beginning at line 775 preserves the full policy matrices, freezes

preservation and exploratory roles separately, and writes held-out rows without altering the model dispatch path or default routing mode.

Contextual query facets and real-head audit. `src/prahf/query_facets.py:101--183` forms overlapping windows only from states captured in one complete contextual prompt pass and retains the exact final-token global row. Lines 193–247 normalize and batch all facet–parent semantic products; lines 251–337 pool valid native K tokens by parent with `index_add_`, enforce the 16:8 GQA map, and compute the full facet–query–head–parent tensor. Lines 340–388 deduplicate per-head nominations before one deterministic final Top- B . The runner at `experiments/paper2_5_iterative_pra/run_query_entry_facets.py:673--958` reproduces the old root, selects window/head policies only on validation identities, records winning span/head provenance and cost, and runs the three exact synthetic controls. The capture at `precompute_query_entry_features.py:42--191` asserts bit-exact final-token parity and records zero independent window forwards. The confirmation at `run_query_entry_propagation.py:74--443` changes only root scores before invoking the previously frozen monotonic native-rank policy; its final parent budget remains unchanged.

Reproduction. The experiment directory is `experiments/paper2_5_iterative_pra/`. Its Gate-1 runner isolates projection correction. Gate 2 captures one 256-token contextual parent pass and derives 32-token local means. Gate 3 uses `precompute_native_qk_features.py` followed by `run_gate3_native_qk_closure.py`. The 620,984,671-byte feature cache is regenerable and excluded from Git; its public manifest pins Qwen revision `c1899de289a04d12100db370d81485cdf75e47ca`. Its SHA-256 is

`edc589cc7886ba843ee598e89e6dab97e500084df8b7ee8dc8ab44ea75993ae8`.

Exact commands are listed in the experiment `README.md`; raw rows, aggregate tables, plots, and graph JSONL are under the corresponding shared result tree.

B Validation Gates

The test suite checks $D = 1$ equivalence to Top- B , deterministic batches, exact synthetic two-hop recovery, deduplication and cycles, depth/beam/unique budgets, all path and frontier modes, zero limits, layer locality, projection-contract enforcement, exact local-bridge recovery, parent deduplication, GQA head mapping, max and Top-4 native reductions, semantic narrowing, threshold caps, exact pre-RoPE Q/K capture, CPU/CUDA identity parity, fixed-identity payload parity, and the no-materialization-before-closure boundary. Gate-3 generation validates every saved graph against the backward-compatible schema-v2 extension and records zero K/V materializations during closure. The feature manifest verifies the frozen model revision, layer 27, 16/8 grouped heads, native head width 128, and no memory-use adapter. The focused iterative/oracle/native/adaptive suite passes 69 tests; the query-facet/head module adds nine focused unit tests for contextual windows, one-pass derivation, exact baseline scoring, GQA mapping, provenance, deduplication, budget conservation, and oracle metrics. All 160 saved Gate-3 graphs and 5,600 result rows satisfy schema, parent-budget, and no-closure-materialization checks. The additive diagnostic preserves 4,000 convergence rows and 340 raw edge-score rows, then adds 2,488 root-hit classifications, 17,710 score-family observations, 2,619 fixed-union calibration candidates, and 780 held-out control rows. Tests cover canonical-oracle identity, annotation groups, displacement, protected-root budgeting,

score summaries, ranking ties, validation leakage, deterministic policy fitting, and cost accounting. The monotonic gate adds 19,200 root-policy rows, 14,080 transition rows, and 1,040 held-out frozen-policy rows; every row conserves its parent budget and retains all locked identities. The entry gate adds 1,792 root rows and 450 four-method confirmation rows. Its manifest records 32 one-pass contextual query captures, zero independent window passes, and zero maximum final-token baseline error; all three synthetic controls pass. The complete repository suite passes 412 tests (one upstream deprecation warning).

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